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Research Directions

My research focuses on **LLM Agent Optimization**, with the goal of building adaptive agents that learn from experience and interaction to perform reliably in complex, evolving real-life environments. Specifically, my work spans three areas:

- **Knowledge-Augmented Reasoning:** GraphRAG [6, 7, 8].
- **Agent Harness Optimization:** Instruction Optimization [5, 11]; Task-Aware Routing [4, 10, 11].
- **Agent Policy Optimization:** Data-Centric Reinforcement Learning [2, 3]; Agentic RL with Simulated Users [14, 15].

I am specifically interested in the topics of **Agent Uncertainty** and **Interactive Intent Alignment**: when agents face real users' incomplete or misleading requests and evolving goals in long-horizon agentic tasks, how can we enable them to recognize their own uncertainty, understand the limits of their capabilities, and remain aligned with users through effective and efficient interaction, such as seeking clarification? Specifically, my work covers:

- **Agent Uncertainty** [12]: Uncertainty Quantification [3]; Uncertainty Calibration [9]; Uncertainty Utilization [2, 3].
- **Interactive Intent Alignment:** Benchmarking Intent Misalignment and Shifting [1, 13]; Interactive Policy Learning [14, 15].

Professional Experience

ByteDance, Inc. *Seattle, WA*
Research Scientist - AML Ark US (Intern) *Jul. 2026 – Oct. 2026*
- Research on modeling interactive alignment tasks under miscommunication and evolving user intent on both public benchmarks and ArkClaw production data [1].
- Work on error attribution analysis on agentic sessions with 1000+ steps and session environment reconstruction.

Amazon, Inc. *Seattle, WA*
Applied Scientist - AWS Connect (Intern) *Apr. 2026 – Jul. 2026*
- Research on Recoverability-Aware Intervention Learning for Dynamic Training-Time Rollout Allocation [2].

Applied Scientist - BRP (Part-time) *Aug. 2025 – Dec. 2025*
- Research on uncertainty-aware reinforcement learning and LLM agents for coding and data analytical tasks.

Applied Scientist - BRP (Intern) *May 2025 – Aug. 2025*
- Research on LLM agents for code generation. Develop MAPRO [5], a self-evolving prompt optimization framework for multi-agent systems. MAPRO was also accepted to AMLC Spotlight and CSS Oral (Amazon Internal Conference).
- Both the coding agents and the MAPRO framework are deployed into the *AIDE (Assistant for Intelligent Deep divE)* project, which aims to build an expert system for deep-dive investigations.

IBM Research *Yorktown Heights, NY*
Student Researcher (Part-time) *Jun. 2024 – May 2025*
- Research on interpretable and explainable foundation models using graph-based retrieval augmented generation in health care, leading to two publications at ACL 2025 and KDD 2025 [6, 7]. See the collaboration project website.

Coleridge Initiative, Inc. *New York, NY*
Research Scientist *Oct. 2021 – Mar. 2023*
- Lead the research project *Food for Thought*, using ML methods to link food and nutrition databases on a secure platform; responsible for its toolkit, evaluation pipelines and cross-team communication (see competition blog).
- Lead the *Unemployment Insurance* project, developing a real-time system to monitor unemployment trends for IL, IN and TN. Lead the Data Team, responsible for all incoming data ingestion and ETL pipelines.

Associated Research Scientist *Aug. 2020 – Oct. 2021*
- Responsible for the *Rich Context* project, extracting dataset mentions from full-text social science publications.
- Host the Kaggle competition *Show US the Data*, responsible for its benchmark and technical implementation. A paper is accepted to Harvard Data Science Review.

Junior Research Scientist (Intern) *Oct. 2019 – Aug. 2020*
- Work on KG completion and text entity extraction for a large-scale knowledge graph in the *Rich Context* competition.

Selected Publications

* indicates equal contribution. To see the full publication list, please visit my Google Scholar here.

[1] **Z. Zhang**, M. Chao, K. Xiao, Z. Chen, D. Zhang, Y. Zhang, Y. Ye, W. Xu. Beyond Oracle Communication: Benchmarking Interactive Intent Alignment Under Miscommunication and Evolving User Intent, 2026, *arXiv*.

[2] **Z. Zhang**, M. Mao, H. Wang, Z. Wang, S. Koelle, J. Yuan, Y. Lin, J. Feng, N. L. Kuang, Y. Ye, W. Niu. Optimizing What Policies Learn From: Recoverability-aware Rollout Intervention Learning, 2026, *arXiv*.

[3] **Z. Zhang**, K. Shi, H. Bao, Z. Wang, T. Ma, Y. Ye. Regulating What Policies Learn From: Geometry-aware Calibrated Policy Optimization, 2026, *arXiv*.

[4] **Z. Zhang***, K. Shi*, Z. Yuan, Z. Wang, T. Ma, K. Murugesan, V. Galassi, C. Zhang, Y. Ye. AgentRouter: A Knowledge-Graph-Guided LLM Router for Collaborative Multi-Agent Question Answering, 2026, *ACL Main*.

[5] **Z. Zhang**, L. Ge, H. Li, W. Zhu, C. Zhang, Y. Ye. MAPRO: Recasting Multi-Agent Prompt Optimization as Maximum a Posteriori Inference, 2026, *EACL Findings*.

[6] **Z. Zhang***, Y. Li*, N. H. L. Le*, Z. Wang, T. Ma, V. Galassi, K. Murugesan, N. Moniz, W. Geyer, N. V. Chawla, C. Zhang, Y. Ye. NGQA: A Nutritional Graph Question Answering Benchmark for Personalized Health-aware Nutritional Reasoning, 2025, *ACL Main*.

[7] **Z. Zhang**, Z. Wang, T. Ma, V. S. Taneja, S. Nelson, N. H. L. Le, K. Murugesan, M. Ju, N. V. Chawla, C. Zhang, Y. Ye. MOPI-HFRS: A Multi-objective Personalized Health-aware Food Recommendation System with LLM-enhanced Interpretation, 2025, *KDD*.

[8] **Z. Zhang***, Z. Wang*, S. Hou, E. Hall, L. Bachman, V. Galassi, J. White, N. V. Chawla, C. Zhang, Y. Ye. Diet-ODIN: A Novel Framework for Opioid Misuse Detection with Interpretable Dietary Patterns, 2024, *KDD*.

[9] K. Shi*, **Z. Zhang***, Y. Ye. SAGE: Answer-Conditioned Uncertainty Targets for Verbal Uncertainty Alignment, 2026, *EMNLP Findings*.

[10] K. Shi*, **Z. Zhang***, Z. Yuan, K. Murugesan, V. Galassi, C. Zhang, Y. Ye. NG-Router: Graph-Supervised Multi-Agent Collaboration for Nutrition Question Answering, 2026, *EACL Main*.

[11] J. Huang*, **Z. Zhang***, K. Shi, Y. Ye, C. Zhang. EvolveRouter: Co-Evolving Routing and Prompt for Multi-Agent Question Answering, 2026, *arXiv*.

[12] K. Shi*, **Z. Zhang***, et al. Uncertainty in LLM Agents: A Comprehensive Survey, 2026, *Preprint*.

[13] H. Bao, **Z. Zhang**, P. Jing, Z. Yuan, K. Shi, Y. Ye. Drift-Bench: Diagnosing Cooperative Breakdowns in LLM Agents under Input Faults via Multi-Turn Interaction, 2026, *ICML*.

[14] J. Huang, **Z. Zhang**, Z. Li, T. Ma, M. Li, Y. Ye, C. Zhang. CLEAT: Learning Value-Guided Clarification for Interactive Agents, 2026, *arXiv*.

[15] H. Bao, **Z. Zhang**, K. Shi, Y. Ye. Not Every User Says It All: Training Tool-Using Agents Beyond Ideal Simulated Users, 2026, *arXiv*.

Education

University of Notre Dame — *South Bend, IN*
Ph.D. in Computer Science — *Jul. 2023 – Present*

New York University, M.S. — *Aug. 2019 – Aug. 2020*

Wuhan University, B.S. — *Sept. 2015 – Jun. 2019*

- **Awards:** 1st Class Scholarship 2016–2018, Leijun Scholarship 2018

Service

- **Conference Senior Area Chair:** KDD'26
- **Conference Reviewer:** NeurIPS, ICML, ICLR, KDD, WWW, ARR.
- **Journal Reviewer:** Transactions on Big Data, Data Mining and Knowledge Discovery, TIST, TMLR, TWEB, PLOS Digital Health.
- **Teaching Assistant:** FA23-CSE-40113, SP24-CSE-40113: Design/Analysis of Algorithms

Invited Talks

- MOPI-HFRS (Sept. 2024) @ IBM Research, Thomas J. Watson Research Center, Yorktown Heights.
- LLM Agents for Code Generation (Jun. 2025) @ Amazon, Seattle.